

Comparison of Additive and FM Synthesis for Piano and Flute Tone Modelling

1 Student Information

- **Full Name:** Zhan Xu
- **Student ID (SID):** 550439255
- **GitHub Username:** zhxu0930-cmd
- **GitHub Repository:** <https://github.com/zhxu0930-cmd/elec5305-project-550439255>
- **GitHub Project Site:** <https://zhxu0930-cmd.github.io/elec5305-project-550439255/>

2 Project Overview

This project will compare additive synthesis and frequency modulation (FM) synthesis for modelling piano and flute tones. The aim is to investigate how effectively the two methods can reproduce important spectral and temporal characteristics of real musical instrument sounds.

Real recordings of piano and flute playing three target notes, C4, E4, and A4, will be analysed in MATLAB. Corresponding tones will then be generated using additive and FM synthesis. The synthesized signals will be compared with the reference recordings using harmonic structure, spectrograms, and spectral similarity measures.

The main research question is:

How effectively can additive synthesis and FM synthesis reproduce the spectral and temporal characteristics of piano and flute tones, and how do the two methods differ in modelling accuracy and parameter complexity?

The project will also investigate whether adding a time-varying amplitude envelope improves the synthesized tones, particularly for piano sounds with strong attack and decay characteristics.

3 Background and Motivation

Musical instruments can produce the same fundamental frequency while sounding very different because of differences in spectral and temporal characteristics, which play an important role in musical timbre [1].

Additive synthesis generates complex tones by summing sinusoidal components with different frequencies and amplitudes. It provides direct control over harmonic structure but may require many parameters. Analysis-based additive synthesis has been used to reconstruct musical instrument tones using time-varying amplitude and frequency information [2]. FM synthesis instead creates complex spectra by modulating the frequency of a carrier signal. It can generate

rich and dynamically varying spectra using a relatively small number of control parameters [3], although its spectral components are less directly controlled.

Piano and flute are selected because they have different sound characteristics. Flute tones are generally more stable, while piano tones have a strong attack and gradual decay. Comparing both instruments provides a useful way to evaluate the strengths and limitations of the two synthesis methods.

The motivation of this project is to examine the trade-off between synthesis accuracy and implementation complexity, rather than simply generating sounds that are perceptually acceptable.

4 Proposed Methodology

MATLAB will be used as the main implementation platform. Reference recordings of piano and flute tones at C4, E4, and A4 will be obtained primarily from the University of Iowa Musical Instrument Samples (MIS) database or other publicly available isolated-instrument datasets where necessary. All recordings will be converted to a common sampling rate and normalised before analysis.

For each reference tone, the waveform, frequency spectrum, and spectrogram will be analysed to identify the fundamental frequency, dominant harmonics, and temporal envelope. The spectral analysis will follow the general concept of sinusoidal and spectral modelling [4]. These characteristics will guide the synthesis parameters.

In additive synthesis, each tone will be generated by summing sinusoidal components corresponding to the fundamental frequency and selected harmonics. Their amplitudes will be estimated from the reference spectrum. In FM synthesis, the carrier frequency, modulation frequency, and modulation index will be adjusted to produce a spectrum similar to the reference tone.

The two methods will be applied to the same six target tones. Their outputs will then be compared with the corresponding real recordings using harmonic structure, spectrograms, and log-spectral distance (LSD). A further modification will add a time-varying amplitude envelope to improve temporal modelling, especially for piano tones.

5 Expected Outcomes

The project is expected to produce a MATLAB-based implementation of both additive and FM synthesis for piano and flute tone modelling. Synthesized tones for C4, E4, and A4 will be generated and compared with real instrument recordings.

The evaluation will include waveform and spectrogram comparisons, harmonic structure analysis, and log-spectral distance (LSD) between the synthesized and reference tones. Lower LSD values will indicate greater spectral similarity. A time-varying amplitude envelope will also be tested as an improvement for modelling the temporal characteristics of the instruments.

The final deliverables will include MATLAB code, comparison results, representative figures and audio examples, together with GitHub documentation and a short project demonstration.

6 Timeline (Weeks 1–13)

Week	Task
1–2	Finalise project topic and research question
3–5	Literature review and collection of piano and flute recordings
6–7	Analyse spectral and temporal characteristics of selected tones
8–9	Implement additive and FM synthesis in MATLAB
10–11	Compare synthesis methods and test amplitude-envelope improvement
12–13	Analyse results, complete final report, and update GitHub documentation

7 References

- [1] J. M. Grey, “Multidimensional Perceptual Scaling of Musical Timbres,” *The Journal of the Acoustical Society of America*, vol. 61, no. 5, pp. 1270–1277, 1977.
- [2] J. M. Grey and J. A. Moorer, “Perceptual Evaluations of Synthesized Musical Instrument Tones,” *The Journal of the Acoustical Society of America*, vol. 62, no. 2, pp. 454–462, 1977.
- [3] J. M. Chowning, “The Synthesis of Complex Audio Spectra by Means of Frequency Modulation,” *Journal of the Audio Engineering Society*, vol. 21, no. 7, pp. 526–534, 1973.
- [4] X. Serra and J. Smith, “Spectral Modeling Synthesis: A Sound Analysis/Synthesis System Based on a Deterministic Plus Stochastic Decomposition,” *Computer Music Journal*, vol. 14, no. 4, pp. 12–24, 1990.